



Quant Week 2

2/6/25





Agenda



1. Icebreaker Question
2. Brainteasers
3. Market Making, continued
4. Market Making Game
5. Project



Ice Breaker

What is the worst class you've ever taken?





Daily Sequence

27, 90, 9, 30, 3, 10, ?

This is from an Optiver OA!



Brain Teaser - Chameleon Colors



A remote island has chameleons that can change between 3 different colors: red, green and blue

When two chameleons meet, they will change their color to the third color (example red and blue ones meet, they become green)

If there are 13 red, 15 green, and 17 blue chameleons, (13,15,17) is it ever possible for all the chameleons to be the same color?

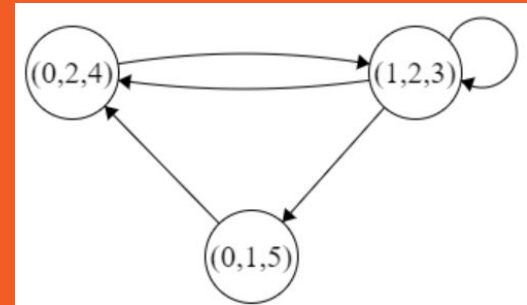


Brain Teaser - Answer

This problem can be reduced to 0 red, 2 green, and 4 blue chameleons (0,2,4).

This is because any (1,1,1) can become (0,0,3)

These are the possible states from (0,2,4):



The combination (1,2,3) can be further reduced to (0,1,2), which can never reach (3,0,0). (0,1,2) will repeat, and will always return to (0,1,2)

Market Making





Order Matching System



- The system that exchanges use to match bids and asks
- All electronic exchanges rely on these systems, also known as matching engines
- Matching engines are common topics in interviews
- Some firms (IMC) may ask you to implement a basic matching engine
- Two states
 - Continuous: Orders matched immediately
 - Auction: Orders matched at fixed intervals, often is used at the start of a day when orders have built up.
- Implementation of these generally uses a priority queue/stack
- Note: Sellers determine the ask price, which represents the minimum price at which they are willing to sell. If a buyer is willing to meet or exceed this price, the trade takes place at the seller's ask price.

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.03	150
	100.02	100
	100.01	50
	100	
60	99.99	
120	99.98	
180	99.97	

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.03	150
	100.02	100 (SOLD)
	100.01	50 (SOLD)
	100	
60	99.99	
120	99.98	
180	99.97	

Order Book Example



Bid (Volume)	Price	Ask (Volume)
	100.05	250
	100.04	200
	100.03	150
	100.02	
	100.01	
	100.00	
60	99.99	

Market Making Game





Game

- Similar to Jane Street Estimathon, or Optiver's market making game
- A question will appear on screen, and one person will make a market
- Everyone must then either buy or sell on that market
- The role of market maker will rotate such that everyone is the market maker once



1: The Population of Montenegro



2: Exponential of the EECS
281 late withdrawal rate



3: The Sum of the Class
Numbers that Toland and I
are taking



4: The amount of people
living in South Quad divided
by π



5: The area of Morocco (sq mi)



6: The percentage on
Patrick's phone right now
times 50



7: The FY23 Michigan Football Revenue



8: The number of unread
messages on Toland's
Messages app



10: The Number of Cars
registered in NYC



9: The Volume of Lake Michigan in cubic miles



11: The Number of Cars
registered in NYC



Project Discussion/Work Time





New Member Form

[Link to Onboarding Document, Slack
and Team Folder](#)